

## BLAKELY, GA, USA

WMO: 720257

Lat: 31.397N

Lon: 84.895W

Elev: 214

StdP: 14.58

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63835

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 27.9       | 31.7 | 14.3                          | 11.4 | 38.2 | 18.9 | 14.3 | 39.2 | 16.9                  | 56.6 | 15.0 | 54.3 | 2.1                   | 340  | 0.329 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.5                   | 97.1            | 76.5 | 95.0 | 76.3 | 92.8 | 76.0 | 81.5                | 88.0 | 80.5 | 87.8 | 79.3 | 87.5 | 4.6                  | 260  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 80.6                            | 160.2 | 84.1 | 78.9 | 151.3 | 83.9 | 77.2 | 142.7 | 83.1 | 45.3          | 88.6 | 44.3 | 88.1 | 43.0 | 87.2 | 86.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 13.4              | 11.6 | 10.3 | DB                         | 22.8 | 98.6               | 4.5 | 1.9                                                | 19.6 | 100.0      | 17.0 | 101.1      | 14.5 | 102.2      | 11.2 | 103.6 |
|                   |      |      | WB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  | N/A   |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 67.4   | 50.6  | 54.1  | 59.6  | 66.5  | 74.0  | 80.7  | 82.3  | 81.8  | 77.6  | 68.9  | 57.6  | 54.3  |
|                                                                        | DBStd       | 13.23  | 9.90  | 8.72  | 8.22  | 5.94  | 5.53  | 3.12  | 3.15  | 3.52  | 4.45  | 7.38  | 7.98  | 8.99  |
|                                                                        | HDD50       | 258    | 114   | 53    | 16    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 19    | 56    |
|                                                                        | HDD65       | 1684   | 457   | 318   | 207   | 51    | 8     | 0     | 0     | 0     | 1     | 47    | 246   | 349   |
|                                                                        | CDD50       | 6610   | 132   | 167   | 314   | 495   | 743   | 922   | 1000  | 987   | 827   | 585   | 248   | 190   |
|                                                                        | CDD65       | 2560   | 10    | 12    | 40    | 96    | 285   | 472   | 535   | 522   | 378   | 167   | 25    | 18    |
|                                                                        | CDH74       | 25291  | 43    | 98    | 394   | 1068  | 2871  | 4805  | 5651  | 5227  | 3329  | 1491  | 224   | 90    |
|                                                                        | CDH80       | 11116  | 1     | 9     | 68    | 274   | 1201  | 2332  | 2742  | 2466  | 1496  | 486   | 36    | 5     |
| (14) Wind                                                              | WSAvg       | 2.9    | 3.6   | 3.6   | 3.8   | 3.3   | 2.9   | 2.5   | 2.3   | 2.1   | 2.6   | 2.6   | 2.7   | 3.0   |
| (15) Precipitation                                                     | PrecAvg     | 56.4   | 6.6   | 6.0   | 5.9   | 4.0   | 3.8   | 5.2   | 5.9   | 4.4   | 3.1   | 2.3   | 3.6   | 5.2   |
|                                                                        | PrecMax     | 79.3   | 13.0  | 10.8  | 12.7  | 9.3   | 8.1   | 13.1  | 12.3  | 9.9   | 9.1   | 8.9   | 9.4   | 9.6   |
|                                                                        | PrecMin     | 37.1   | 1.0   | 0.9   | 0.1   | 0.4   | 0.3   | 0.8   | 1.2   | 1.7   | 0.2   | 0.0   | 0.9   | 1.1   |
|                                                                        | PrecStd     | 10.2   | 3.0   | 2.5   | 2.7   | 2.6   | 2.1   | 3.0   | 2.6   | 2.0   | 2.3   | 2.2   | 2.0   | 2.6   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 78.8  | 81.4  | 85.7  | 89.0  | 94.7  | 99.2  | 98.7  | 99.5  | 96.9  | 90.8  | 84.0  | 81.0  |
|                                                                        |             | MCWB   | 67.8  | 67.4  | 65.3  | 69.8  | 71.6  | 76.2  | 76.7  | 80.2  | 74.0  | 72.7  | 69.0  | 70.4  |
|                                                                        | 2%          | DB     | 73.4  | 77.1  | 81.5  | 86.1  | 91.1  | 96.9  | 96.5  | 96.7  | 93.1  | 88.3  | 79.5  | 76.9  |
|                                                                        |             | MCWB   | 65.5  | 64.0  | 65.8  | 68.9  | 72.1  | 75.6  | 77.4  | 77.9  | 73.9  | 71.7  | 66.6  | 68.2  |
|                                                                        | 5%          | DB     | 70.3  | 72.9  | 79.0  | 82.4  | 90.0  | 93.5  | 93.8  | 93.3  | 90.8  | 85.9  | 76.6  | 72.5  |
|                                                                        |             | MCWB   | 64.4  | 62.6  | 64.2  | 67.4  | 71.5  | 75.5  | 77.4  | 77.7  | 73.5  | 70.8  | 65.6  | 67.4  |
|                                                                        | 10%         | DB     | 66.2  | 70.1  | 74.8  | 80.9  | 87.7  | 91.0  | 91.2  | 90.9  | 88.5  | 81.9  | 72.5  | 70.0  |
|                                                                        |             | MCWB   | 61.5  | 61.5  | 62.4  | 66.0  | 71.4  | 75.0  | 77.2  | 77.3  | 73.7  | 69.5  | 63.8  | 65.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 71.1  | 70.4  | 71.7  | 75.0  | 78.6  | 81.4  | 82.5  | 84.2  | 80.8  | 78.9  | 75.0  | 73.4  |
|                                                                        |             | MCDB   | 76.3  | 75.1  | 77.3  | 81.6  | 82.7  | 86.1  | 90.5  | 90.9  | 83.8  | 85.1  | 79.2  | 77.3  |
|                                                                        | 2%          | WB     | 68.7  | 67.9  | 69.3  | 73.2  | 76.4  | 79.7  | 81.1  | 82.1  | 78.6  | 76.8  | 71.6  | 71.1  |
|                                                                        |             | MCDB   | 72.2  | 74.8  | 75.0  | 79.9  | 83.7  | 87.9  | 88.8  | 89.3  | 85.5  | 83.4  | 75.9  | 74.4  |
|                                                                        | 5%          | WB     | 65.8  | 65.8  | 67.6  | 71.6  | 75.1  | 78.5  | 80.0  | 81.0  | 77.5  | 74.4  | 69.0  | 68.9  |
|                                                                        |             | MCDB   | 69.8  | 72.3  | 75.2  | 78.6  | 83.1  | 87.5  | 88.3  | 88.2  | 85.8  | 80.6  | 73.8  | 72.4  |
|                                                                        | 10%         | WB     | 62.5  | 63.2  | 65.7  | 69.7  | 73.8  | 77.6  | 79.0  | 79.7  | 76.2  | 72.3  | 66.0  | 65.6  |
|                                                                        |             | MCDB   | 66.6  | 67.3  | 73.5  | 76.4  | 82.5  | 87.1  | 87.7  | 87.6  | 84.8  | 79.3  | 72.3  | 69.1  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 20.5  | 22.8  | 24.0  | 24.6  | 23.2  | 20.6  | 19.5  | 19.1  | 20.3  | 23.2  | 24.1  | 20.8  |
|                                                                        |             | MCDBR  | 22.8  | 26.2  | 26.4  | 27.3  | 26.1  | 24.7  | 21.3  | 21.7  | 23.8  | 24.7  | 27.0  | 22.1  |
|                                                                        |             | MCWBR  | 14.4  | 15.1  | 12.9  | 11.8  | 9.8   | 7.5   | 6.9   | 7.3   | 8.1   | 10.1  | 14.6  | 15.2  |
|                                                                        | 5% WB       | MCDBR  | 18.8  | 21.2  | 20.8  | 20.6  | 20.5  | 20.2  | 18.8  | 19.1  | 19.0  | 18.7  | 19.3  | 18.4  |
|                                                                        |             | MCWBR  | 14.1  | 15.2  | 12.1  | 11.9  | 9.2   | 7.7   | 7.6   | 7.6   | 7.9   | 8.8   | 12.9  | 14.2  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.326 | 0.333 | 0.355 | 0.384 | 0.434 | 0.477 | 0.509 | 0.500 | 0.426 | 0.373 | 0.351 | 0.331 |
|                                                                        | taud        |        | 2.515 | 2.489 | 2.440 | 2.362 | 2.252 | 2.196 | 2.116 | 2.139 | 2.328 | 2.450 | 2.455 | 2.516 |
|                                                                        | Ebn at Noon |        | 286   | 294   | 294   | 288   | 273   | 260   | 251   | 252   | 268   | 276   | 274   | 278   |
|                                                                        | Edh at Noon |        | 28    | 31    | 35    | 39    | 44    | 47    | 50    | 49    | 39    | 32    | 29    | 26    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 953   | 1134  | 1512  | 1859  | 2040  | 1936  | 1911  | 1759  | 1581  | 1365  | 1069  | 828   |
|                                                                        | RadStd      |        | 58    | 136   | 125   | 149   | 143   | 144   | 116   | 117   | 126   | 143   | 87    | 83    |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (42 neighbors) | +0.83 | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +288  | +187  |

Nomenclature: See separate page